

MacroEnergy.jl: A large-scale multi-sector energy system framework

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Summary

MacroEnergy.jl (aka Macro) is an open-source framework for multi-sector capacity expansion modeling and analysis of macro-energy systems (Levi et al., 2019). It is written in Julia (Bezanson et al., 2017) and uses the JuMP (Dunning et al., 2017) package to interface with a wide range of mathematical solvers. It enables researchers and practitioners to design and analyze energy and industrial systems that span electricity, fuels, bioenergy, steel, chemicals, and other sectors. The framework is organized around a small set of sector-agnostic components that can be combined into flexible graph structures, making it straightforward to extend to new technologies, policies, and commodities. Its companion packages support decomposition methods and other advanced techniques, allowing users to scale models across fine temporal and spatial resolutions. MacroEnergy.jl provides a versatile platform for studying energy transitions at the detail and scale demanded by modern research and policy.

Statement of Need

The increasing complexity of energy systems necessitates advanced modeling tools to support decision-making in infrastructure planning, R&D decisions and policy design. This complexity comes from the challenge of ensuring the reliability of grids with large amounts of renewable generation and storage, increased coupling and electrification of energy-intensive sectors, greater diversity in the technologies and policies being deployed, and many other factors.

Capacity expansion modelling frameworks have improved substantially in recent years. A wider range of problems can now be solved thanks to improvements in the underlying formulations and solvers while access to richer data sources has enabled more realistic representations of resources, weather and demand. Looking ahead, further improvements are on the horizon, including non-linear technology formulations that capture richer trade-offs (Fälth et al., 2023; Heo & Macdonald, 2024; Levin et al., 2023), tighter integration with integrated assessment models and other tools (Gong et al., 2023; Gøtske et al., 2025; Odenweller et al., 2025), and novel approaches to scaling up problem size (Liu et al., 2024; Parolin et al., 2025; Pecci & Jenkins, 2025).

There has also been some convergence in the design and capabilities of modelling frameworks as the field comes to understand what is required to produce robust, policy-relevant results. Recent studies suggest that capacity expansion models must consider decades of operational data (Ruggles et al., 2024; Ruhnau & Qvist, 2022), may require temporal resolution as fine as five minutes (Levin et al., 2024; Mallapragada et al., 2018), and should capture spatial heterogeneity at the county level (Frysztacki et al., 2023; Krishnan & Cole, 2016; Qiu et al., 2024; Serpe et al., 2025). In addition, they must be able to represent a wide variety of coupled sectors as the majority of emission reductions will come from outside the electricity

sector. Electricity-centric frameworks; such as PyPSA (T. Brown et al., 2017), GenX (Jenkins & Sepulveda, 2017), Calliope (Pfenninger & Pickering, 2018), and others (Blair et al., 2014; P. Brown et al., n.d.; He et al., 2024; Howells et al., 2011); developed the computational capabilities needed to optimize grids over long time series of hourly or sub-hourly data in order to properly incorporate variable renewable energy generation and storage. In recent years, several have begun to extending their frameworks to include other sectors, such as hydrogen, fuels, and industrial processes. On the other hand, economy-wide models; such as TIMES (Loulou et al., 2005), TEMOA (Hunter et al., 2013) and others; have long been able to represent multiple sectors through the use of flexible graph-based structures. However, they do not have the computational performance required to include long, high-resolution time series.

Extending existing models to new sectors or to dramatically improve performance often requires rewriting core routines or layering new modules on top. This complicates validation, obscures interactions across the system, and leaves the codebase hard to maintain. In the authors' experience from previous development, the frameworks remain architected around their original sectors, making it problematic to exclude those sectors and quickly increasing the difficulty and time required to add new features.

MacroEnergy.jl was designed to overcome these limitations. Its architecture is based on a small set of sector-agnostic components that can be combined into graphs to represent networks, technologies, and policies in any sector. Features are largely independent of one another, allowing users to focus on how best to represent their technology or policy of interest instead of working around the existing code.

MacroEnergy.jl is also designed from the ground-up to scale to large, multi-sector problems. Modeling across coupled sectors greatly increases runtimes, often making problems intractable (Parolin et al., 2025). Techniques such as model compression and the use of representative periods can ease the computational burden, but eventually large-scale models reach the limits of what can be solved on a single computing node. To scale further, methods which allow models to be solved across computing clusters are essential. MacroEnergy.jl was designed with these challenges in mind. Its data structures and graph-based representation of energy systems enable sectoral, temporal and spatial decompositions by default. It also includes a suite of companion packages, which provide advanced decomposition algorithms (Pecci et al., 2025), automatic model scaling (Macdonald, 2024), and example systems (Macdonald et al., 2025). Other companion packages are under development. These will provide representative period selection and other tools to enhance MacroEnergy.jl. MacroEnergy.jl and its companion packages are registered Julia packages and are freely available on GitHub or through the Julia package manager.

Use Cases

MacroEnergy.jl can be used to optimize the design and operation of energy and industrial systems, investigate the value of new technologies or policies, optimize investments in an energy system over multiple years, and many other tasks. It is being used for several ongoing investigations of regional energy systems, including as part of the Net-Zero X Global Initiative - a research consortium involving top research institutions around the world developing shared modeling methods and completing detailed, actionable country-specific studies supporting net-zero transitions.

The framework was designed with three user profiles in mind. Where possible, we have passed modelling complexity upstream to developers, so that most users can build and run models faster and with less coding knowledge.

- Users: Want to create and optimize a real-world system using MacroEnergy.jl. They should be able to do this with little or no coding, and without knowledge of MacroEnergy.jl's components or internal structure.

- 91 ▪ Modelers: Want to add new assets, sectors, or public policies to MacroEnergy.jl. They
- 92 will need to be able to code in Julia and understand some of MacroEnergy.jl's components,
- 93 but they do not require knowledge of its internal structure or underlying packages.
- 94 ▪ Developers: Want to change or add new features, model formulations or constraints to
- 95 MacroEnergy.jl. They will require detailed knowledge of MacroEnergy.jl's components,
- 96 internal structure, and underlying packages.

97 Structure

98 MacroEnergy.jl models are made up of four core components which are used to describe the
 99 production, transport, storage and consumption of various commodities. The components can
 100 be connected into multi-sectoral networks of commodities. They are commodity-agnostic so
 101 can be used for any flow of a good, energy, etc. While we believe MacroEnergy.jl will most
 102 often be used to study energy systems, commodities can also be data, money, or more abstract
 103 flows.

104 The four core components are:

- 105 1. Edges: describe and constrain the flow of a commodity
- 106 2. Nodes: balance flows of one commodity and allow for exogenous flows into and out of a
- 107 model. These can be used to represent exogenous demand or supply of a commodity.
- 108 3. Storage: allow for a commodity to be stored over time.
- 109 4. Transformations: allow for the conversion of one commodity into another by balancing
- 110 flows of one or more commodities.

111 These four core components can be used directly to build models but most users will find
 112 it easier to combine them into Assets and Locations. Assets are collections of components
 113 that represent real-world infrastructure such as power plants, industrial facilities, transmission
 114 lines, etc. For example, a water electrolyzer asset would include edges for electricity and water
 115 inputs and hydrogen output, and a transformation to convert between them. Locations are
 116 collections of Nodes which represent physical places where assets are situated and commodities
 117 can be transported between. While Edges can only connect to Nodes of the same Commodity,
 118 Locations are an abstraction that simplifies the user-input required to connect different
 119 commodities across physical places. Together, Assets and Locations allow for models to be
 120 truer to life and easier to analyze.

121 Assets and Locations in turn form Systems which represent an energy and/or industrial
 122 system. Most often, each System will be optimized separately given a user-defined operating
 123 period. Several Systems can be combined into a Case. Cases can be used for multi-stage
 124 capacity expansion models, rolling-horizon optimization, sensitivity studies, and other work
 125 requiring multiple snapshots or versions of an energy system. MacroEnergy.jl can automatically
 126 manage the running of these different Cases for users, either directly or in combination with
 127 MacroEnergySolver.jl package.

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References

- Bezanson, J., Edelman, A., Karpinski, S., & Shah, V. B. (2017). Julia: A fresh approach to numerical computing. *SIAM Review*, 59(1), 65–98.
- Blair, N., Dobos, A. P., Freeman, J., Neises, T., Wagner, M., Ferguson, T., Gilman, P., & Janzou, S. (2014). *System advisor model, sam 2014.1. 14: General description*. National Renewable Energy Lab.(NREL), Golden, CO (United States).
- Brown, P., Carag, V., Chen, Y., Chernyakhovskiy, I., Cohen, S., Cole, W., Duraes de Faria, V., Gagnon, P., Halloran, C., Hamilton, A., Ho, J., Mindermann, K., Mowers, J., Mowers, M., Obika, K., Pham, A., Schleifer, A., Sergi, B., Serpe, L., ... Vanatta, M. (n.d.). *Regional Energy Deployment System Model 2.0 (ReEDS 2.0)*. <https://www.nrel.gov/analysis/reeds/index.html>
- Brown, T., Hörsch, J., & Schlachtberger, D. (2017). PyPSA: Python for power system analysis. *arXiv Preprint arXiv:1707.09913*.
- Dunning, I., Huchette, J., & Lubin, M. (2017). JuMP: A modeling language for mathematical optimization. *SIAM Review*, 59(2), 295–320.
- Fälth, H. E., Mattsson, N., Reichenberg, L., & Hedenus, F. (2023). Trade-offs between aggregated and turbine-level representations of hydropower in optimization models. *Renewable and Sustainable Energy Reviews*, 183, 113406.
- Frysztacki, M. M., Hagenmeyer, V., & Brown, T. (2023). Inverse methods: How feasible are spatially low-resolved capacity expansion modelling results when disaggregated at high spatial resolution? *Energy*, 281, 128133.
- Gong, C. C., Ueckerdt, F., Pietzcker, R., Odenweller, A., Schill, W.-P., Kittel, M., & Luderer, G. (2023). Bidirectional coupling of the long-term integrated assessment model REgional model of INvestments and development (REMIND) v3. 0.0 with the hourly power sector model dispatch and investment evaluation tool with endogenous renewables (DIETER) v1. 0.2. *Geoscientific Model Development*, 16(17), 4977–5033.
- Gøtske, E. K., Pratama, Y., Andresen, G. B., Gidden, M. J., Victoria, M., & Zakeri, B. (2025). First steps towards bridging integrated assessment modeling and high-resolution energy system models: A scenario matrix for a low-emissions sector-coupled european energy system. *Environmental Research Communications*, 7(8), 085010.
- He, G., Mallapragada, D., Macdonald, R., Law, J., Shaker, Y., Zhang, Y., Cybulsky, A., Chakraborty, S., & Giovanniello, M. (2024). *DOLPHYN: Decision optimization for low-carbon power and hydrogen networks*. Github.
- Heo, T., & Macdonald, R. (2024). Effects of charging and discharging capabilities on trade-offs between model accuracy and computational efficiency in pumped thermal electricity storage. *arXiv Preprint arXiv:2411.07805*.
- Howells, M., Rogner, H., Strachan, N., Heaps, C., Huntington, H., Kypreos, S., Hughes, A., Silveira, S., DeCarolis, J., Bazillian, M., & others. (2011). OSeMOSYS: The open source energy modeling system: An introduction to its ethos, structure and development. *Energy Policy*, 39(10), 5850–5870.
- Hunter, K., Sreepathi, S., & DeCarolis, J. F. (2013). Modeling for insight using tools for energy model optimization and analysis (temoa). *Energy Economics*, 40, 339–349.
- Jenkins, J. D., & Sepulveda, N. A. (2017). *Enhanced decision support for a changing electricity landscape: The GenX configurable electricity resource capacity expansion model*.
- Krishnan, V., & Cole, W. (2016). Evaluating the value of high spatial resolution in national capacity expansion models using ReEDS. *2016 IEEE Power and Energy Society General*

- 181 Meeting (PESGM), 1–5.
- 182 Levi, P. J., Kurland, S. D., Carbajales-Dale, M., Weyant, J. P., Brandt, A. R., & Benson, S.
183 M. (2019). Macro-energy systems: Toward a new discipline. *Joule*, 3(10), 2282–2286.
- 184 Levin, T., Bistline, J., Sioshansi, R., Cole, W. J., Kwon, J., Burger, S. P., Crabtree, G. W.,
185 Jenkins, J. D., O’Neil, R., Korpås, M., & others. (2023). Energy storage solutions to
186 decarbonize electricity through enhanced capacity expansion modelling. *Nature Energy*,
187 8(11), 1199–1208.
- 188 Levin, T., Blaisdell-Pijuan, P. L., Kwon, J., & Mann, W. N. (2024). High temporal resolution
189 generation expansion planning for the clean energy transition. *Renewable and Sustainable*
190 *Energy Transition*, 5, 100072.
- 191 Liu, B., Bissuel, C., Courtot, F., Gicquel, C., & Quadri, D. (2024). A generalized benders
192 decomposition approach for the optimal design of a local multi-energy system. *European*
193 *Journal of Operational Research*, 318(1), 43–54.
- 194 Loulou, R., Remme, U., Kanudia, A., Lehtila, A., & Goldstein, G. (2005). Documentation for
195 the times model part ii. *Energy Technology Systems Analysis Programme*, 384.
- 196 Macdonald, R. (2024). *MacroEnergyScaling.jl*. Github.
- 197 Macdonald, R., Pecci, F., Li, Anna, Lyu, R., & Atouife, M. (2025). *MacroEnergyExamples.jl*.
198 Github.
- 199 Mallapragada, D. S., Papageorgiou, D. J., Venkatesh, A., Lara, C. L., & Grossmann, I. E.
200 (2018). Impact of model resolution on scenario outcomes for electricity sector system
201 expansion. *Energy*, 163, 1231–1244.
- 202 Odenweller, A., Ueckerdt, F., Hampp, J., Ramirez, I., Schreyer, F., Hasse, R., Muessel, J.,
203 Gong, C. C., Pietzcker, R., Brown, T., & others. (2025). REMIND-PyPSA-eur: Integrating
204 power system flexibility into sector-coupled energy transition pathways. *arXiv Preprint*
205 *arXiv:2510.04388*.
- 206 Parolin, F., Weng, Y., Colbertaldo, P., & Macdonald, R. (2025). Sectoral and spatial
207 decomposition methods for multi-sector capacity expansion models. *arXiv Preprint*
208 *arXiv:2504.08503*.
- 209 Pecci, F., Bonaldo, L., & Jenkins, J. D. (2025). *MacroEnergySolvers.jl*. Github.
- 210 Pecci, F., & Jenkins, J. D. (2025). Regularized benders decomposition for high performance
211 capacity expansion models. *IEEE Transactions on Power Systems*.
- 212 Pfenninger, S., & Pickering, B. (2018). Calliope: A multi-scale energy systems modelling
213 framework. *Journal of Open Source Software*, 3(29), 825.
- 214 Qiu, L., Khorramfar, R., Amin, S., & Howland, M. F. (2024). Decarbonized energy system
215 planning with high-resolution spatial representation of renewables lowers cost. *Cell Reports*
216 *Sustainability*, 1(12).
- 217 Ruggles, T. H., Virguez, E., Reich, N., Dowling, J., Bloomfield, H., Antonini, E. G., Davis, S.
218 J., Lewis, N. S., & Caldeira, K. (2024). Planning reliable wind-and solar-based electricity
219 systems. *Advances in Applied Energy*, 15, 100185.
- 220 Ruhnau, O., & Qvist, S. (2022). Storage requirements in a 100% renewable electricity system:
221 Extreme events and inter-annual variability. *Environmental Research Letters*, 17(4), 044018.
- 222 Serpe, L., Cole, W., Sergi, B., Brown, M., Carag, V., & Karmakar, A. (2025). The importance
223 of spatial resolution in large-scale, long-term planning models. *Applied Energy*, 385, 125534.